



**African Population and
Health Research Center**

Transforming lives in Africa through research.

Introduction to OHDSI & Data Standardization using OMOP Common Data Model

FarajaMH 2026

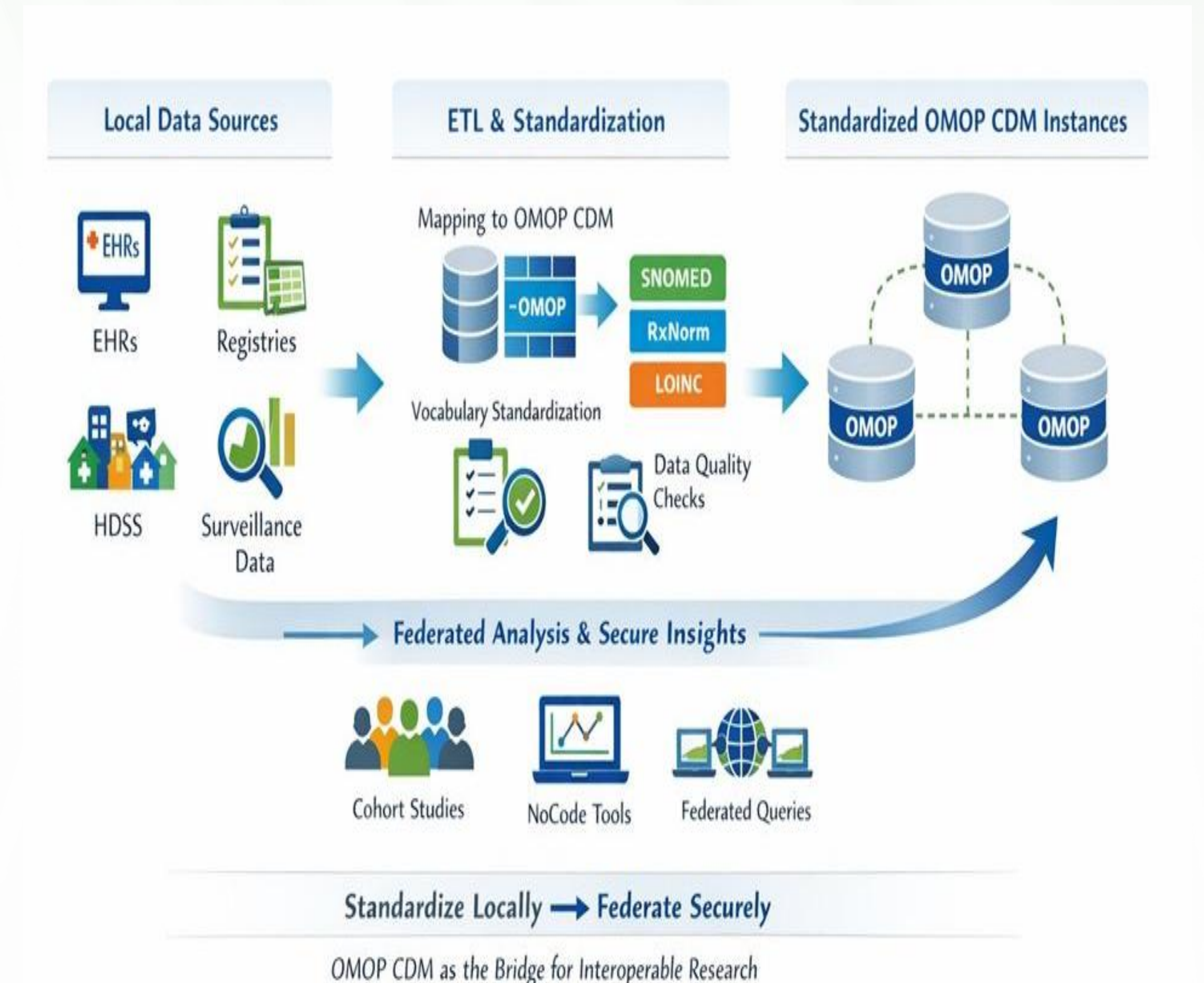
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14TH SEPTEMBER 2026

Content Outline



1	Data Standardization
2	OHDSI
3	OMOP-CDM
4	Standard Vocabularies
5	Software, ETL Pipeline and Analysis Tools
6	Evidence Generation in Africa



1. Data Standardization



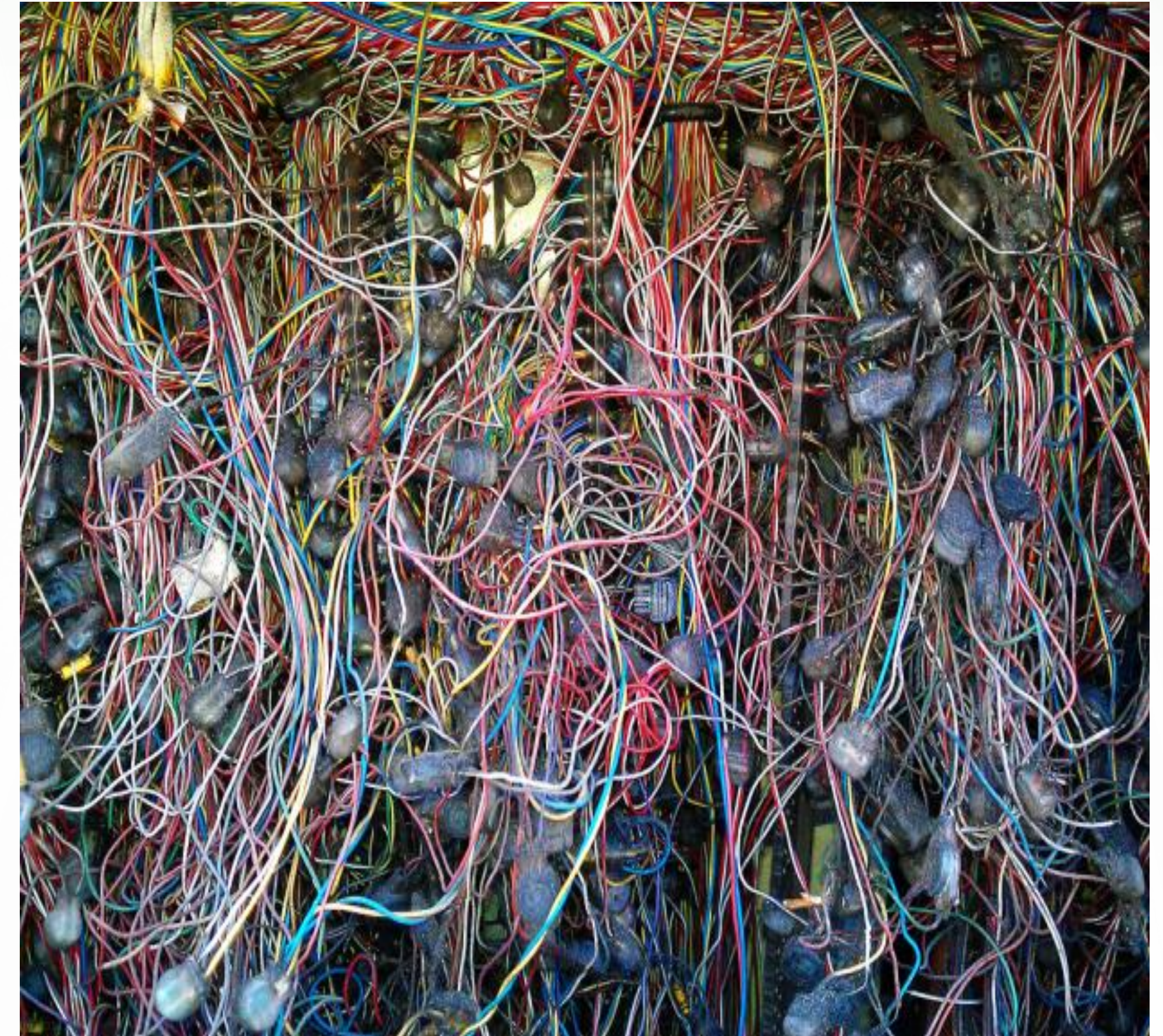
Converting data into a common format to enable users to process and analyze it consistently.

Why is it important?

- ✓ **Consistency:** Ensures data is uniformly formatted, making it easier to compare and analyze across different systems
- ✓ **Accuracy:** Reduces errors and discrepancies by using standardized formats and definitions.
- ✓ **Data Integration:** Facilitates merging data from multiple sources
- ✓ **Supports Analytics and Decision-Making:** High-quality, standardized data enhances the reliability of comparative analytics.

Challenges in Data Standardization

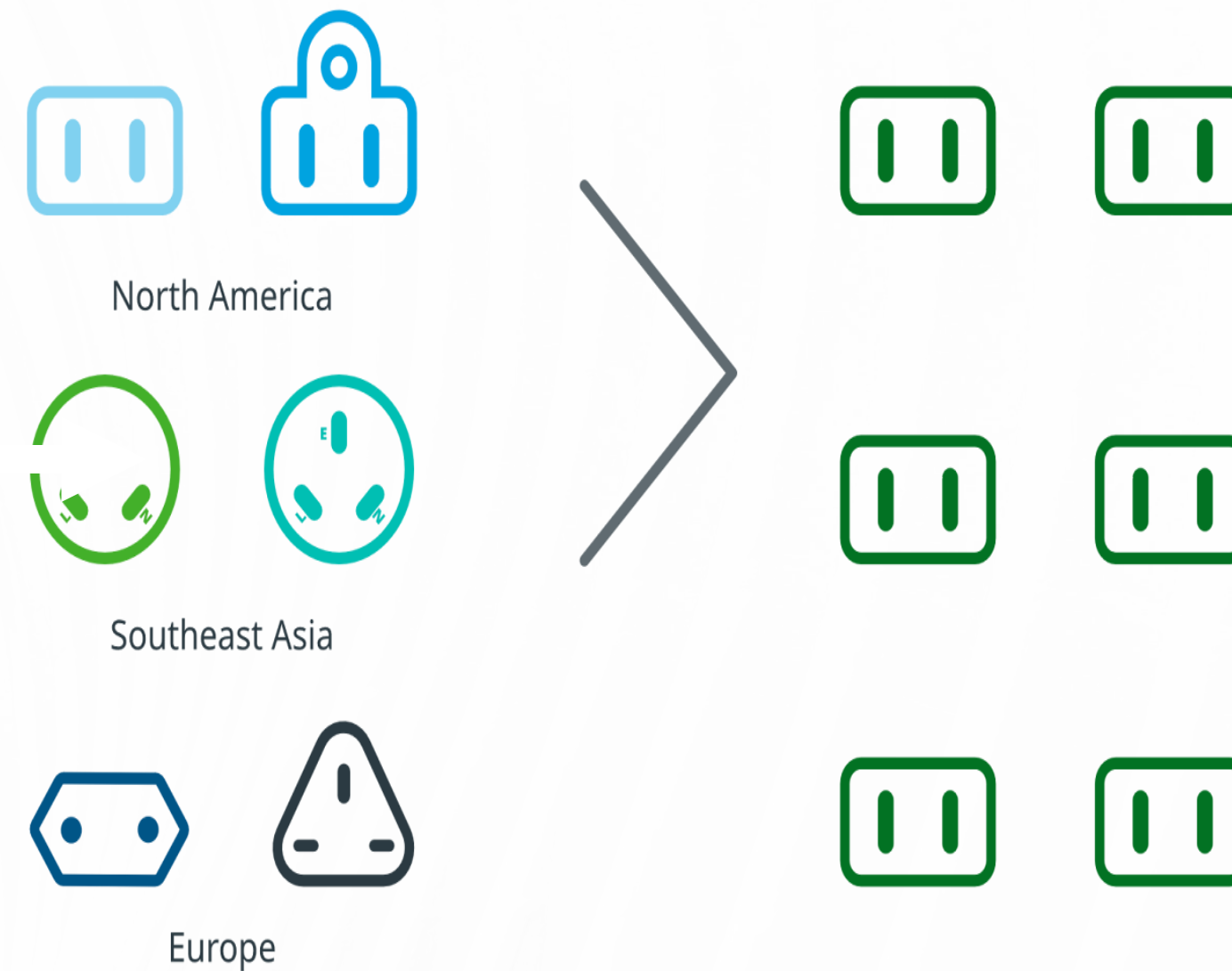
- **Diverse Data Formats:** Different data sources may use various formats (e.g., XLSX, CSV, DTA, SAV, JSON, XML) and structures.
- **Semantic Inconsistency:** Different data collection standards, terminology, coding systems, or abbreviations used.
- **Multilingual Chaos:** Multiple languages (e.g., English, Swahili, French, Spanish, Portuguese, etc.)



Solution to Standardization challenges



A **data model** is a representation of data typically collected about things or events and the relationships between them.

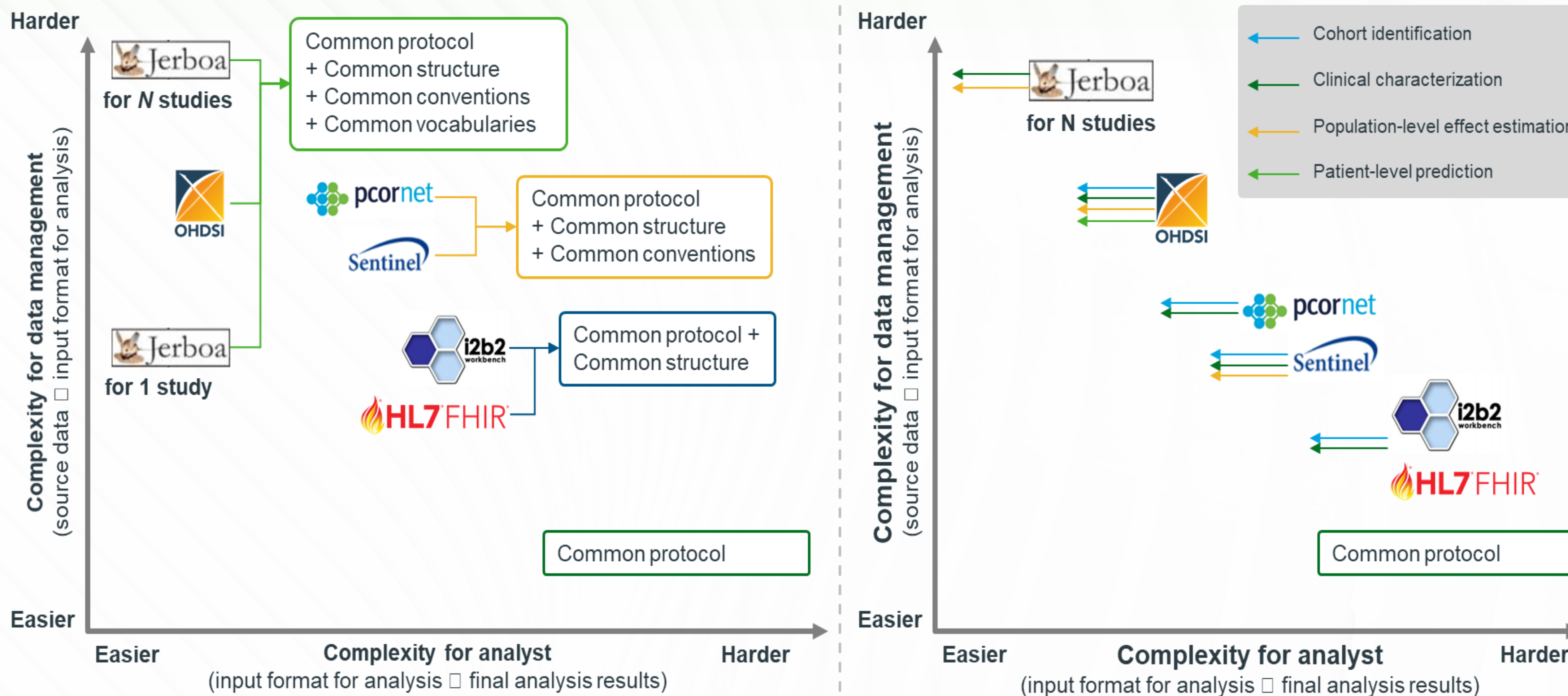


OHDSI standardizes the way research is conducted and these include:

- Standardized vocabularies
- The Observational Medical Outcomes Partnership Common Data Model (OMOP-CDM)
- Standardized analytical methods and tools

Comparison of Common Data Models

Balancing trade-offs in data management vs. analysis complexity



Why OMOP?



1

Open Source

CDM, tools, methods, and documentation all publicly available

2

Standardization

Standard CDM, vocabulary/ontology, tools, methods, data quality, and documentation

3

Research Community

Large research community with multiple stakeholders and disciplines

4

Multi-country/multi-center research

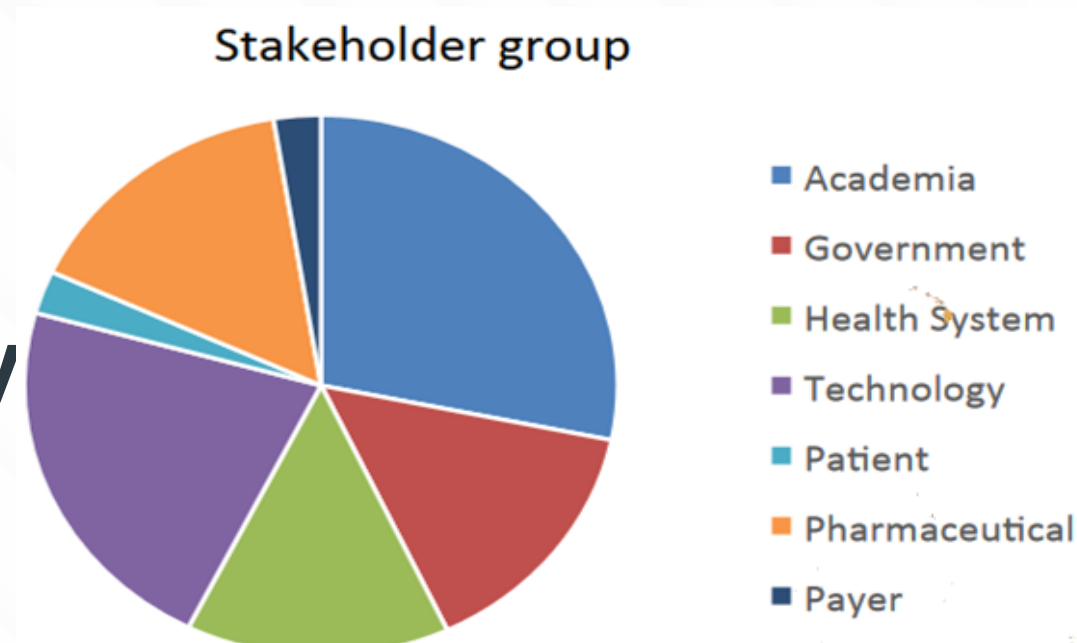
Large scale research using standardized tools and methods

2.



What OHDSI is:

- ✓ Open Source
- ✓ Research Community
- ✓ Data Harmonization



Why Choose OHDSI/OMOP:

- ✓ **Fast, reliable** studies across a series of datasets and data types
- ✓ **Reduced cost of ownership** including understanding coding schemes, writing statistical programs across databases or developing software
- ✓ **Expanded data access** via the OHDSI network and remote multi-center database studies

OHDSI Collaborators:

- 4,751 collaborators
- >1,100 organizations
- 88 countries from 6 continents

OHDSI Network:

- 544 data sources
- 956M unique patient records

Join the Journey: <https://www.ohdsi.org/>

Joining the Journey

- <https://www.ohdsi.org/>
- Whether you're a software developer, physician or clinical researcher, there's a place for everyone in the OHDSI community

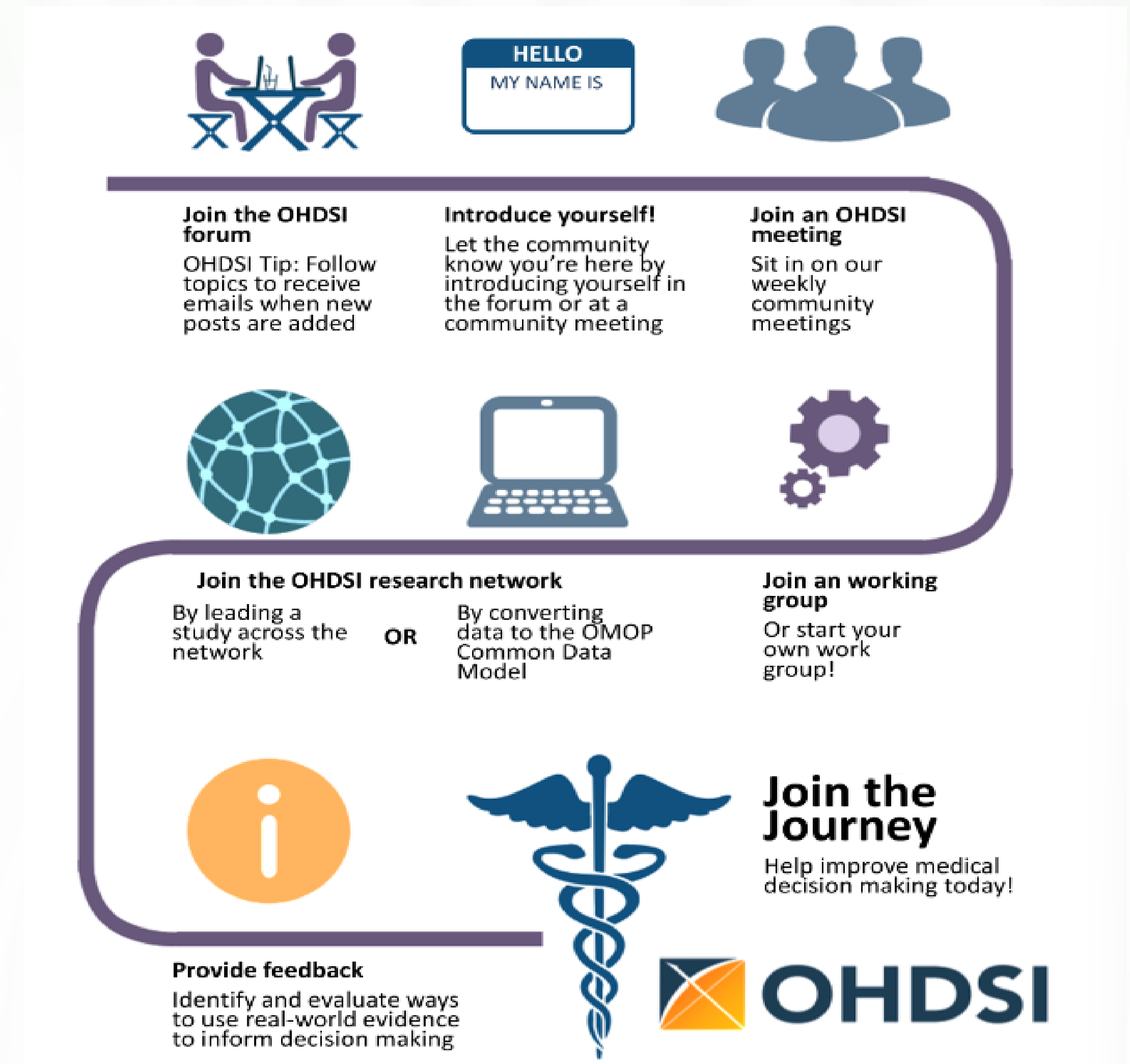
Welcome to OHDSI!

The Observational Health Data Sciences and Informatics (or OHDSI, pronounced "Odyssey") program is a multi-stakeholder, interdisciplinary collaborative to bring out the value of health data through large-scale analytics. All our solutions are open-source.

OHDSI has established an international network of researchers and observational health databases with a central coordinating center housed at Columbia University.

Learn more [about our community](#), about [our mission, vision and values](#), and how you can [help support the OHDSI community](#).

Join the Journey



OHDSI Workgroups & Collaborations

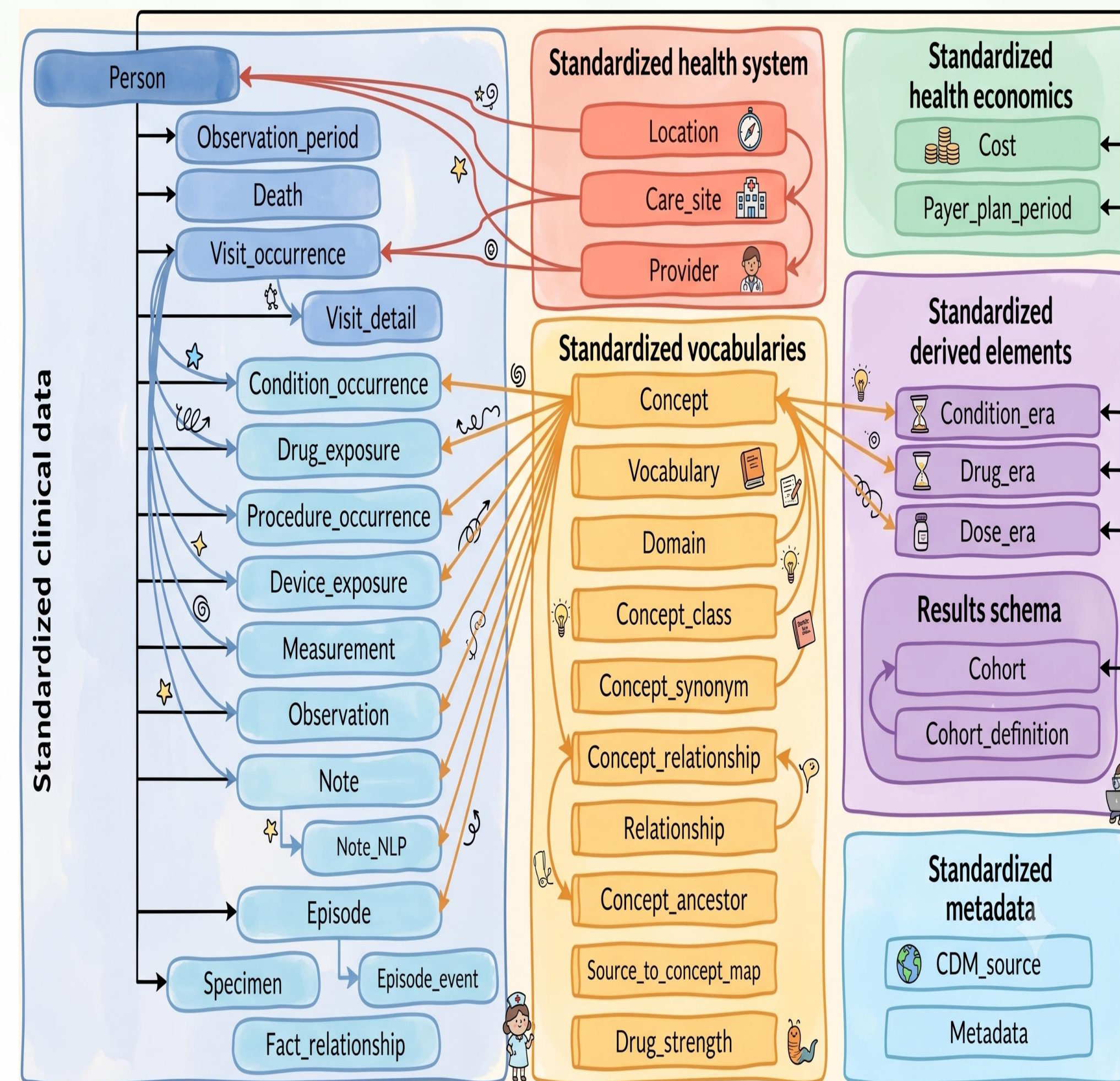


APAC	ATLAS/WebAPI	Clinical Trials	CDM	CDM Vocab Subgroup	Data Network Quality	Dentistry	Education
HADES	Health Equity	Latin America	NLP	Oncology	Registry	Steering Group	Vaccine Vocabulary
Early-Stage Researcher	Eye Care & Vision Research	FHIR and OMOP	GIS	Methods Research	Perinatal & Reproductive Health Group	Psychiatry	Surgery & Perioperative Medicine
	Medical Imaging	Medical Devices	Open-Source Community	Patient-Level Prediction	Healthcare Systems	Phenotype	

Clinical Domains: Type 1 Diabetes Mellitus, Type 2 Diabetes Mellitus, Hypertension, HIV, Depression, Suicide Attempts, Heart Failure, Atrial Fibrillation, Multiple Myeloma, Alzheimer's Disease, Hemorrhagic Events, Prostate Cancer, Breast Cancer, Lung Cancer, Parkinson's Disease, Acute Myocardial Infarction...etc

3. OMOP-CDM

- Relational database model structured using a set of tables and defined relationships between them to standardize diverse healthcare data for large-scale observational research
- Current CDM version is CDM v5.4
- Made up of 39 Tables with predefined fields and data types – (25 to standardize data; 2 for metadata; 10 for standardized vocabularies; 2 for results)



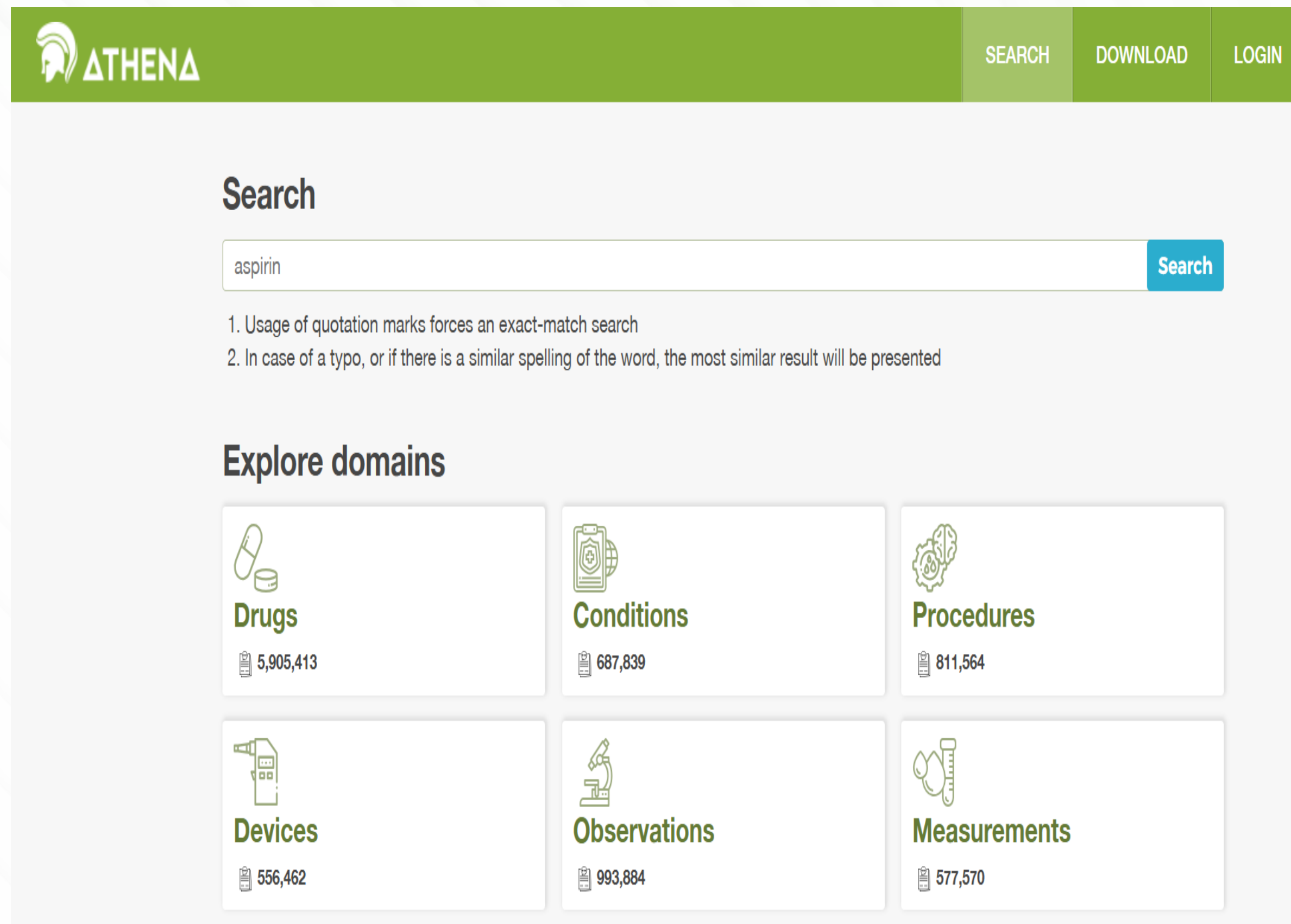
<https://ohdsi.github.io/CommonDataModel/cdm54.html>

4. Standard Vocabularies

- Is a foundation **set of codes or concepts** in OMOP that enables transparent and consistent terminology across observational databases
- Contains a system of vocabularies, Classification, Domains and concepts all consolidated into a common format and **integrated in the OMOP CDM tables**
- Many vocabularies are adopted from national or international organisations, such as ICD, RxNorm, SNOMED-CT, LOINC etc
- It serves to support the OHDSI research community in conducting efficient reproducible observational research
- Supported by OHDSI and **regularly updated, maintained, and improved** by the Odysseus Inc.

- **11,027,290 concepts**
 - 3,598,454 standard concepts
 - 847,008 classification concepts
- **142 vocabularies**
- **44 domains**
- **82,142,038 concept relationships**
- **87,967,689 ancestral relationships**
- **4,673,156 concept synonyms**

Athena Web Application



- Allows you to both search and load standardized vocabularies.
- It is a resource to be used, not a software tool to install
- Download a zip file with all Standardized Vocabularies you need for your OMOP CDM and unzip the package.
- Vocabularies with standard concepts and very common usage are preselected.

<https://www.athena.ohdsi.org/>

5. Software, ETL Pipeline and Analysis Tools



- Scans the source data, providing detailed information on the tables, fields, and values that appear in a field
- Source data can be in comma-separated text files, or in a database (MySQL, SQL Server, Oracle, PostgreSQL, Microsoft APS, Microsoft Access, Amazon RedShift)
- Generates a report that can be used as a reference when designing the ETL



- Designed to read and display the White Rabbit scan document
- Through a graphical user interface to allow a user to connect/Map source data to tables and columns within the OMOP CDM
- Generates documentation for the ETL process, it does not generate code to create an ETL



- Tool to aid the manual process of creating vocabulary mapping.
- Makes suggested mappings based on the textual similarity of code descriptions
- Code connections need to be manually reviewed and approved or Flagged
- Only proposes concepts that are marked as standard concepts in the vocabulary

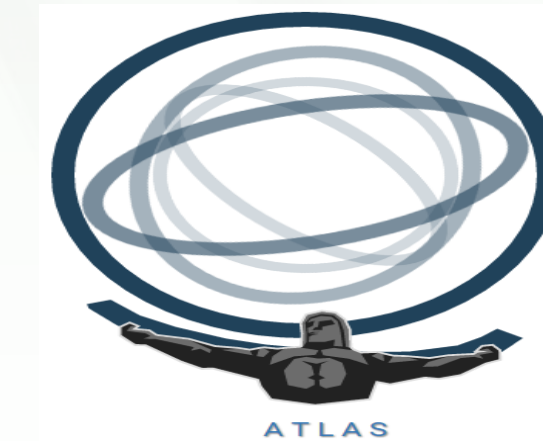
<https://www.ohdsi.org/software-tools/>



- An R package that runs a harmonized data quality assessment terminology to data that has been standardized in the OMOP CDM
- Quality checks are organized according to the Kahn Framework i.e Conformance, Completeness, Plausibility
- For each check, the result is compared to a set threshold



- An R package that provides characterization against an OMOP CDM instance
- Produces a set of results with descriptive statistics stored in a database.
- The stored results can be integrated into ATLAS (a web-based tool) as the “Data Sources” function that enables visualization access.

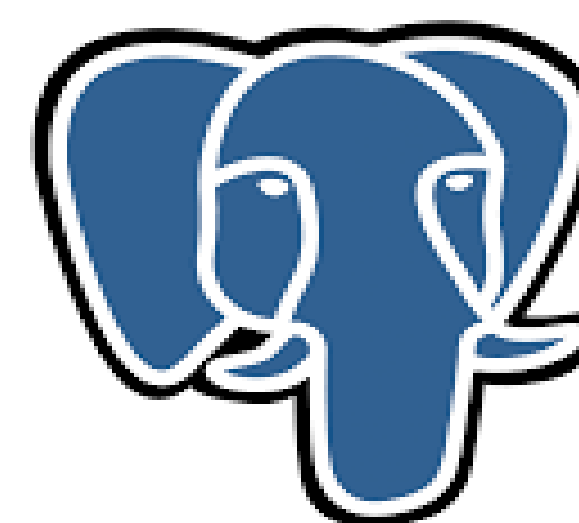


- A web-based tool developed that facilitates the visualization and execution of analyses on standardized, patient-level, observational data in the OMOP-CDM format.
- Cohorts can be created by defining groups of people based on an exposure to a drug or a particular condition
- Patient profiles can be viewed within a specific cohort allowing visualization of a particular subject's records.

<https://www.ohdsi.org/software-tools/>

ETL Implementation Tools

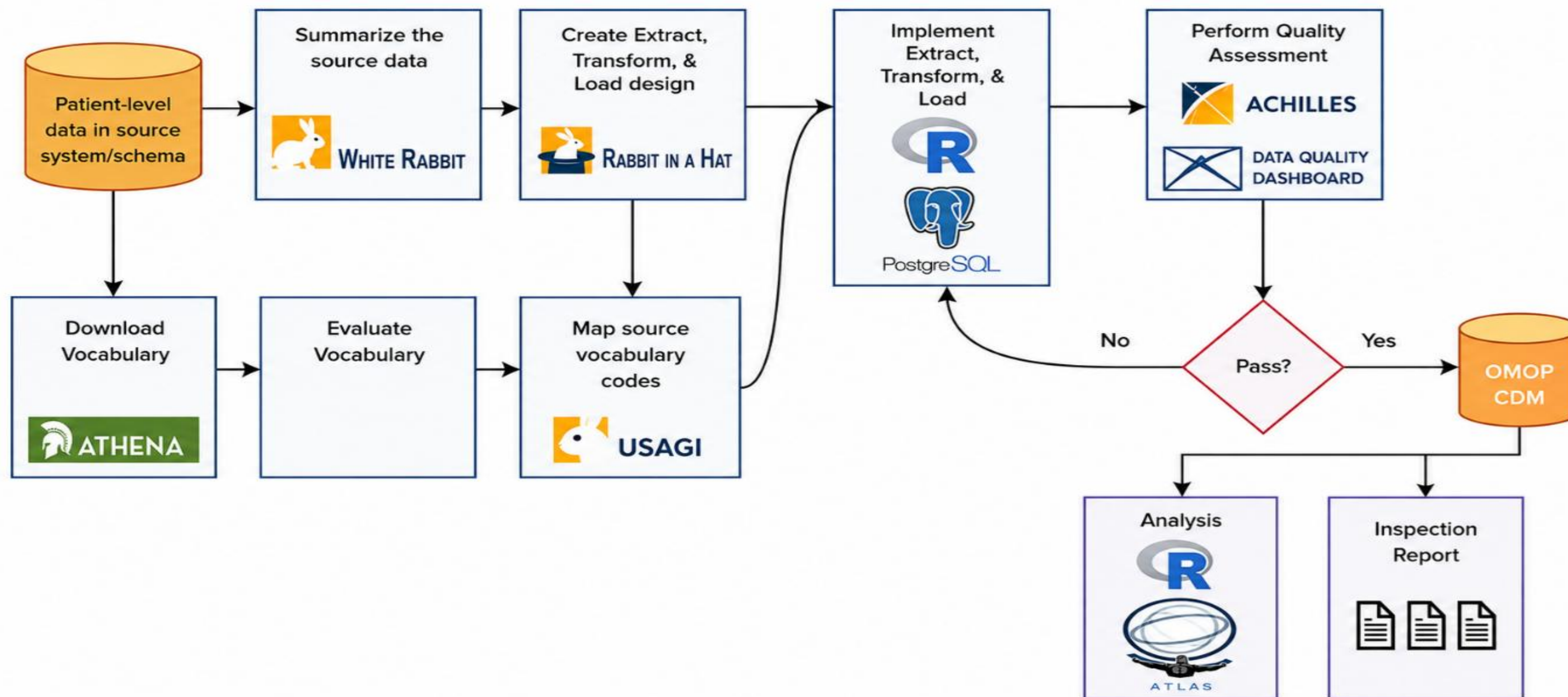
- Once the design and code mappings are completed, the ETL process can be implemented using the below some tools



PostgreSQL



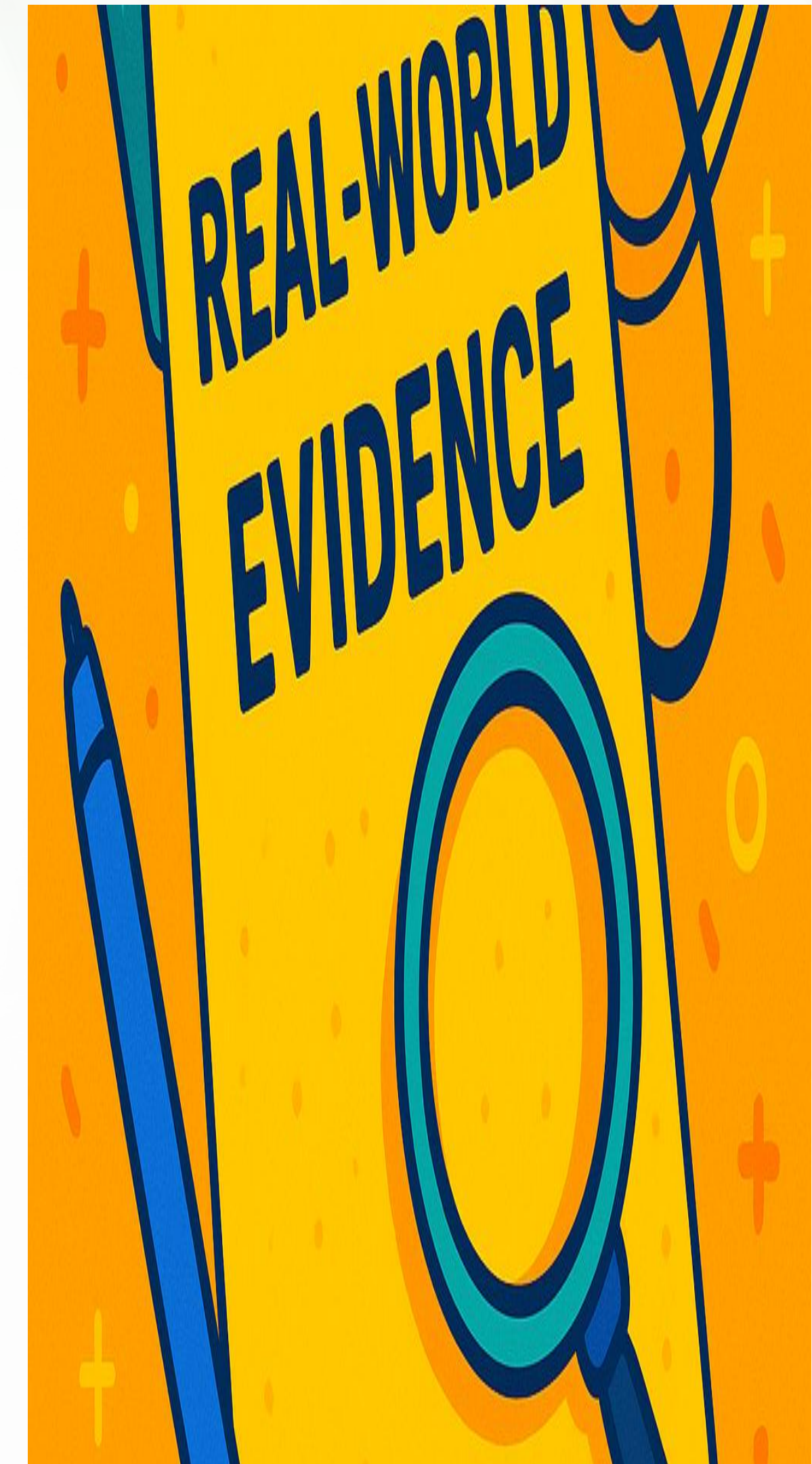
ETL Pipeline



6. Evidence Generation in Africa



- Inspire Mental Health project in APHRC - transformed longitudinal Population data of Primary (Iganga-Mayuge, Kagando, Kilifi HDSS) and Secondary data to OMOP CDM related to depression, Anxiety, psychosis
- INSPIRE 2.0 – successfully transformed Verbal Autopsy data from 6 HDSS sites in Sub-Saharan Africa
- Peach Project in APHRC – transformed COVID-19 data from NUHDSS to OMOP-CDM
- DSWB Project - COVID-19 data from Senegal and HIV data from Cameroon, Ethiopia, Uganda, Kenya transformed to OMOP-CDM
- The LAISDAR project in Rwanda - transformed EHR and COVID-19 data of 14 hospitals to OMOP CDM





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THANK YOU



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